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Roll No

FT-8004 (2) (CBGS)

B.E. VIII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Safety in Rail and Road Transport

Time : Three Hours

Maximum Marks : 70

- Note:** i) Answer any five questions.
ii) All questions carry equal marks.
iii) All parts of each question are to be attempted at one place.
iv) Draw neat and clean diagram.

1. a) Discuss the different types of ballasts with its functions. 7
b) Explain the requirements of sleepers, rails and fastening in railway engineering. 7
2. a) What is coning wheels? What are the advantages and disadvantages? 7
b) Draw the cross-section of a permanent way and briefly state the function of its three important components. 7
3. a) Discuss the types of railways tracks, point and detail of station yard. 7
b) Explain the cause of rail accidents. Write its safety measures taken to minimize the rail accidents. 7
4. a) Explain traffic conflict with their types. 7
b) Draw the cross-section of a typical road on embankment and show different road margins. 7

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5. a) Discuss in detail the vehicular characteristics which affect the highway design and traffic performance. 7
b) Explain in detail the various types of carriageway markings commonly used. 7
6. a) Write the classification of traffic signals and carriage. 7
b) Explain the term traffic volume. What are methods of carrying out traffic volume studies? 7
7. a) Explain compensation of gradient on horizontal curves. 7
b) Define highway intersections and principle of highway lighting. 7
8. a) What are the peculiarities of the corrugation phenomenon of rails? 7
b) What are the transition curves? What type of transition curves are used in the Indian railway? 7
